

Yuel: Improving the Generalizability of Structure-Free Compound–Protein Interaction Prediction

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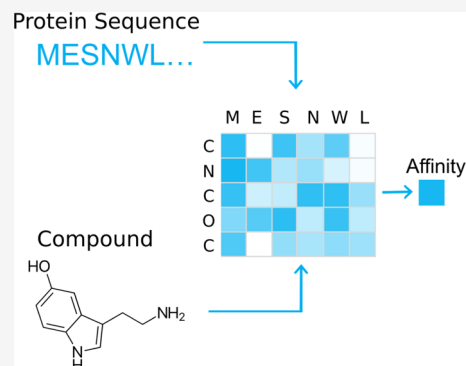


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ABSTRACT: Predicting binding affinities between small molecules and the protein target is at the core of computational drug screening and drug target identification. Deep learning-based approaches have recently been adapted to predict binding affinities and they claim to achieve high prediction accuracy in their tests; we show that these approaches do not generalize, that is, they fail to predict interactions between unknown proteins and unknown small molecules. To address these shortcomings, we develop a new compound–protein interaction predictor, Yuel, which predicts compound–protein interactions with a higher generalizability than the existing methods. Upon comprehensive tests on various data sets, we find that out of all the deep-learning approaches surveyed, Yuel manifests the best ability to predict interactions between unknown compounds and unknown proteins.



INTRODUCTION

Drug discovery¹ is an extremely slow and labor-intensive process, and even approved drugs often suffer from poor activity (rare interactions with the intended protein target) and high side-effect profiles (binding with other targets). Drug screening and drug target identification are two areas in drug development that aim to solve the problems of poor activity and high side effects, respectively. At the core of drug screening and drug target identification is compound–protein binding affinity evaluation. Although various experimental assays² have been developed for biochemical high-throughput drug screening and target identification, they are usually prohibitively resource-intensive and time-consuming. Computational methods³ offer an alternative way due to the low cost and high processing rate. Most computational compound–protein interaction (CPI) evaluation approaches are either structure-based or structure-free (sequence-based). Structure-based methods rely on molecular simulations of small-molecule positioning (molecular docking)^{4–8} in the pocket of the target protein,⁹ and structure-free methods (or cheminformatic methods)¹⁰ are based on prior knowledge of known molecules that interact with the corresponding targets. The advantage of molecular docking¹¹ is that it relies on the physics of interatomic interactions; hence, the approach is independent of prior information of known binders. However, structure-based molecular docking is limited by heavy reliance on high-quality three-dimensional (3D)-structure data of the protein targets, the speed, and deficiencies of the force field. Determining the atomic resolution structures of protein–ligand complexes through currently available experimental techniques, such as X-ray crystallography,¹² nuclear magnetic

resonance (NMR) spectroscopy,¹³ and cryogenic electron microscopy,¹⁴ is time-consuming in practice, resulting in only a limited number of solved structures. In addition, docking and simulation-based methods generally require tremendous computational resources. In contrast to docking, the advantage of cheminformatic approaches is that they offer rapid prediction of small molecules that would bind to a given target if and only if there is prior information about the binding potential of that target. Their limitation is the transferability of information to other targets for novel predictions.

In recent years, deep learning has been used for both structure-based^{15–19} and structure-free^{20–23} methods. Several structure-free deep learning-based neural networks^{20–23} are DeepDTA,²⁰ DeepConv-DTI,²¹ DeepAffinity,²² and multi-objective neural network (MONN).²³ DeepDTA employs convolutional neural network (CNN) blocks to learn representations from the raw protein sequences and SMILES²⁴ strings and combine these representations to feed into a fully connected (FC) layer for CPI prediction. DeepConv-DTI also employs CNN blocks to learn protein features but uses the Morgan/Circular²⁵ drug fingerprint of compounds and analyzes compounds as graphs to learn the features. DeepAffinity utilizes pre-training bidirectional recurrent neural networks (RNNs) to encode compound features and unifies

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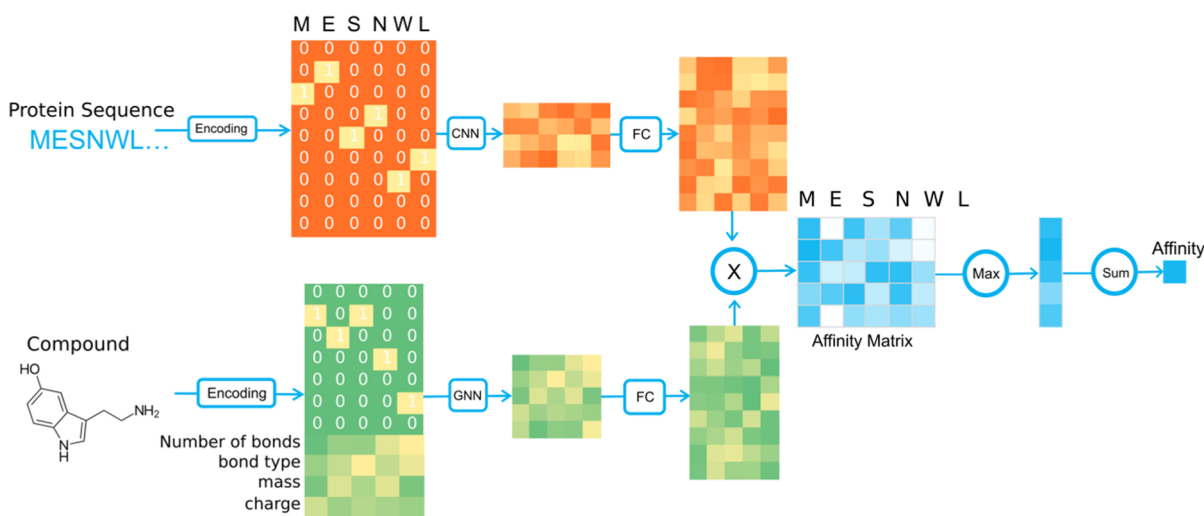


Figure 1. Architecture of Yuel. CNN is the convolution neural network. GNN is the graph neural network, and FC is fully connected layers. The “X” symbol means the outer product of the compound feature matrix and the protein feature matrix.

RNNs with CNNs for CPI predictions. MONN is a multi-objective neural network that encodes compound and protein features with graph neural networks (GNNs) and CNNs and predicts pair-wise non-covalent interactions and binding affinities with an attention-based prediction module. All methods require only two-dimensional (2D) structures of compounds and primary sequences of proteins as inputs. These methods employ widely used deep neural network architectures, such as CNN,²⁶ RNN, and GNN,²⁷ to process and extract contextual features from the input protein sequence and compound 2D structure data. However, the current approaches all follow a common paradigm, which is to encode the compound and protein features first and then concatenate them and subject them to FC layers to predict CPI. This *encoding-FC* paradigm has been successful in many fields;²⁸ however, we demonstrate this by simply conforming to this paradigm, and neural networks are inclined to memorize the protein sequence or the compound structure. In fact, we find that when shuffling the residues in proteins, which should give meaningless results, current neural networks give similar results as on the unshuffled proteins. This overfitting issue significantly limits its generalizability, that is, the ability to predict the interaction between unknown compounds and unknown proteins, thus hindering the use of deep learning techniques in the field of drug virtual screening and target identification. Here, we develop a new CPI predictor, *Yuel* (Figure 1), which predicts CPI with higher generalizability than other methods by carefully designing the architecture components, especially the FC layers. Upon comprehensive tests on various data sets, we find that *Yuel* manifests the best ability to predict the interactions between unknown compounds and unknown proteins. The key architectural component which facilitates *Yuel*'s success is a combination of feature-wise FC layers with an attention-based affinity-prediction module. Additionally, this architectural choice allows *Yuel* to identify the specific atoms in both the compound and the protein which interact with each other.

RESULTS

Preparation of Data Sets. We prepare three data sets, Davis,²⁹ Metz,³⁰ and PDBbind,³¹ to evaluate the ability of *Yuel* to predict CPIs. The Davis data set contains interactions of 72

kinase inhibitors with 442 kinases covering >80% of the human catalytic protein kinome. The Metz data set is collected from more than 150 000 kinase inhibitory values, comprising more than 3800 compounds tested against 172 different protein kinases.³⁰ The PDBbind data set is a high-quality set of protein–ligand complexes with available structural data and the corresponding binding affinities. The PDBbind data set contains structural information, but we only use the sequence of the protein and the SMILES of the compound. We divide each of the three data sets into a training set and a test set at a ratio of 8–2, resulting in three training sets (Davis/train, Metz/train, and PDBbind/train) and three test sets (Davis/test, Metz/test, and PDBbind/test).

The number of compounds, proteins, and CPIs in a training set may affect the training of a model. For example, if a data set has 100 proteins, 1000 compounds, and 1000 CPIs, then each compound uniquely corresponds to a specific CPI. Trained on such a training set, a model may predict the CPI by only using the compound information no matter what the protein is. Thus, we propose a metric, the completeness of a data set $C = N_{\text{interaction}} / (N_{\text{protein}} \cdot N_{\text{compound}})$, to evaluate whether a data set is suitable for training a neural network model. N_{protein} is the number of proteins, N_{compound} is the number of compounds, and $N_{\text{interaction}}$ is the number of interactions in the data set. If the completeness of a data set is 1, it means that the interaction between each compound and each protein is included in the data set. The completeness of Davis, Metz, and PDBbind is 0.35, 0.14, and 0.00052, respectively. A data set with an extremely low completeness is not suitable as a training set but can be used as a test set.

Yuel Predicts Compound–Protein Interactions with Improved Generalizability. We train and test *Yuel* and two other structure-free neural networks, DeepConv-DTI²¹ and DeepDTA,²⁰ on the three data sets, Davis,²⁹ Metz,³⁰ and PDBbind.³¹ In addition, we also build a Null-Protein model and a Null-Compound model. The Null-Protein model ignores the input of the protein sequence (Figure S1), and the Null-Compound model ignores the input of the compound (Figure S2). We first train on the training set and test on the corresponding test set; for example, we train on Davis/train and test on Davis/test. We predict the affinities of all compound–protein pairs in the test set and find that the

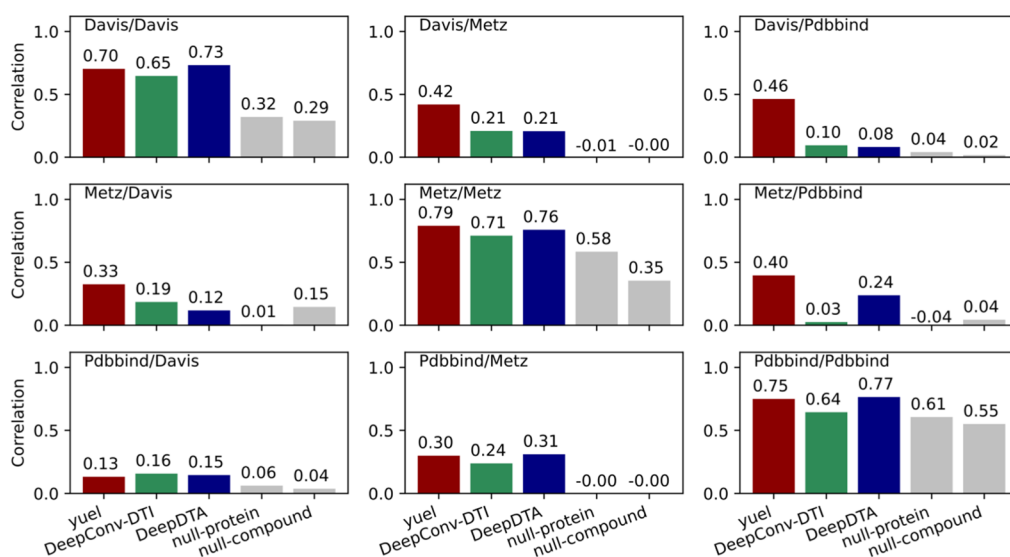


Figure 2. Testing Yuel, DeepConv-DTI, DeepDTA, Null-Protein, and Null-Compound models on different data sets. The label “DS1/DS2” on the upper left corner of each panel means that the models are trained on DS1 and tested on DS2. The Y-axis indicates the Pearson coefficient correlation.

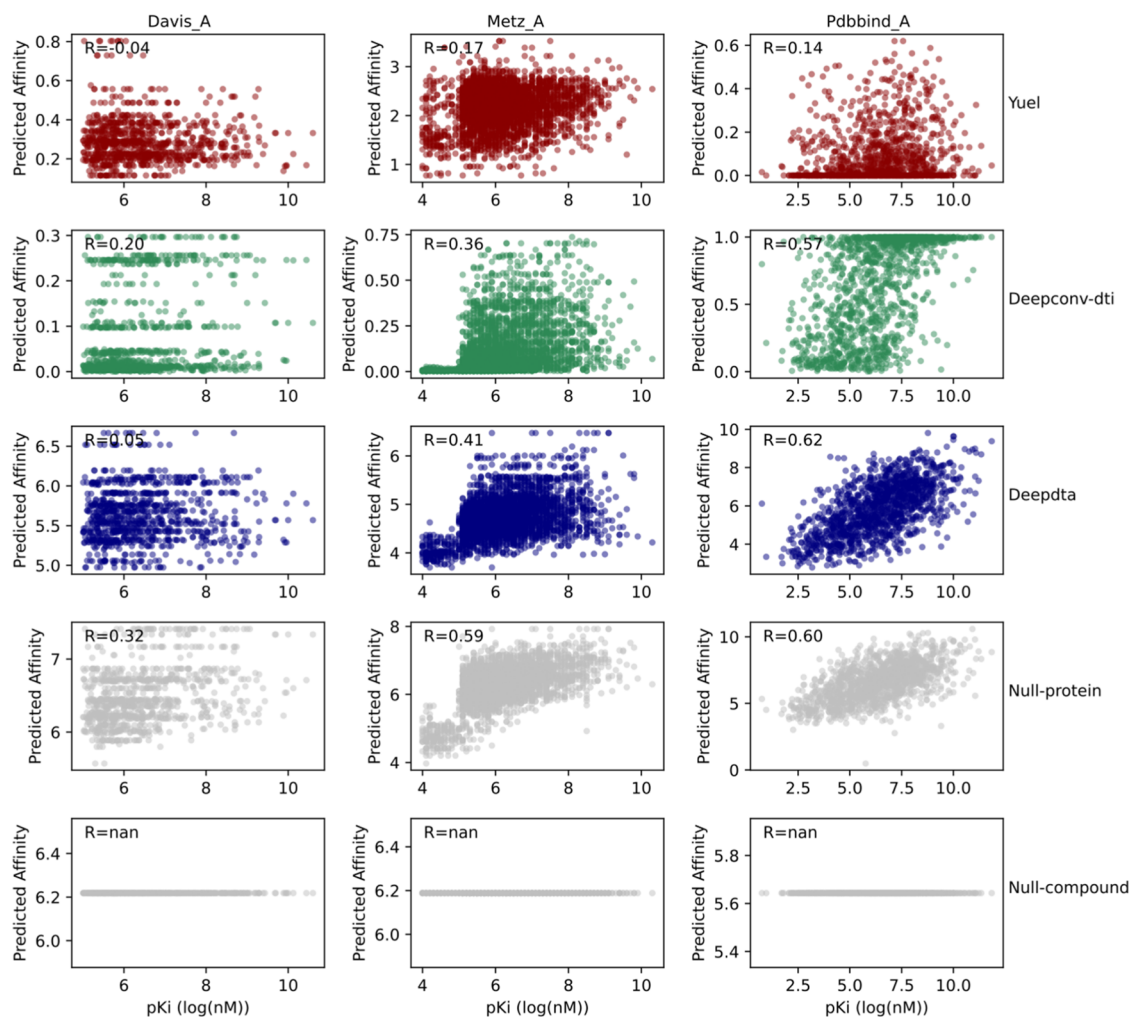


Figure 3. Testing Yuel, DeepConv-DTI, DeepDTA, Null-Protein, and Null-Compound models on the PDBbind (Alanine), Davis (Alanine), and Metz (Alanine) data sets with each protein sequence replaced with a single alanine. The X-axis is the experimental affinity, and the Y-axis is the predicted affinity. The label on the upper left corner is the Pearson correlation coefficient.

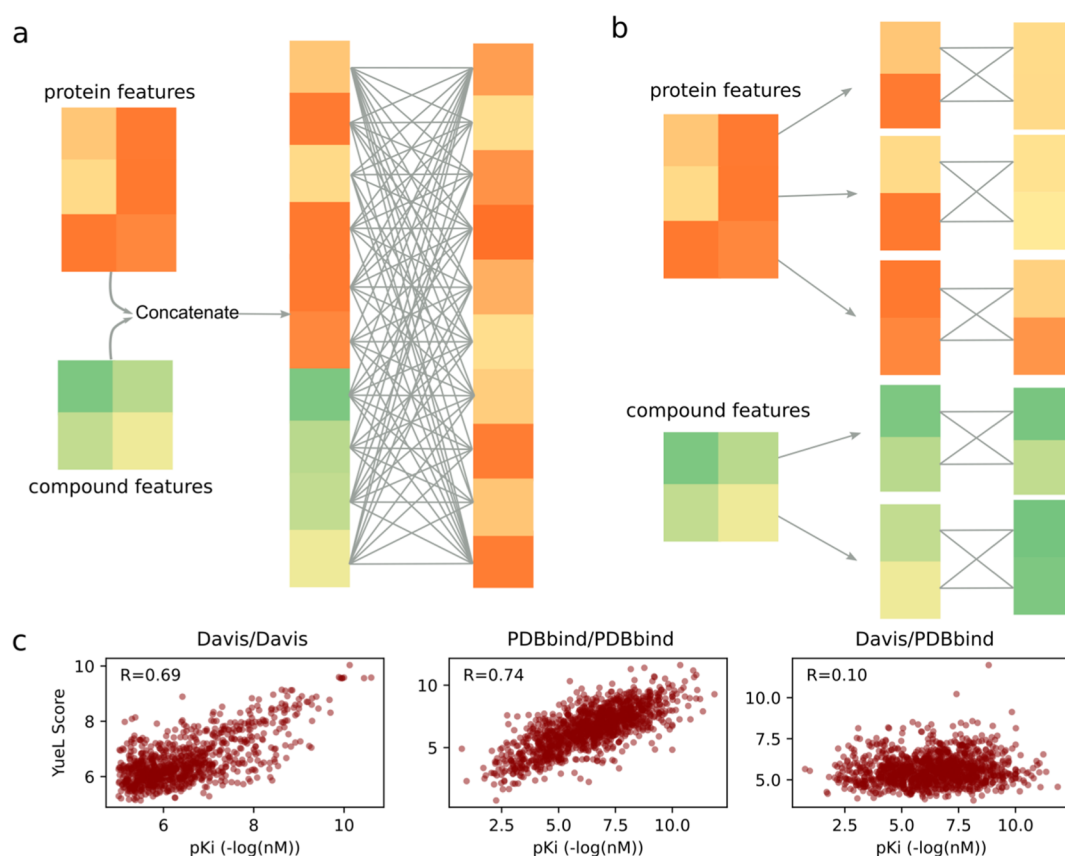


Figure 4. Comparison of position-wise FC layers and feature-wise FC layers. (a) The protein features and the compound features are first concatenated and flattened to a 1-D feature vector. The 1-D feature vector is then subject to FC layers. (b) The protein features and the compound features are first split into individual residue features and atom features. Each residue feature and each atom feature are subject to FC layers individually. Finally, the residue features and the atom features are multiplied to obtain an attention matrix. (c) Testing Yuel-cc on Davis and PDBbind. The label “DS1/DS2” above each panel means that the model is trained on DS1 and tested on DS2.

Pearson correlation coefficients between the experimental affinities and predicted affinities (Figure 2) of Yuel and DeepDTA on Davis, Metz, and PDBbind are all above 0.7, slightly higher than that of DeepConv-DTI. However, this significant result is not surprising as the training and test sets are sourced from the same data set; thus, many proteins and compounds are shared between the training set and the test set. In fact, even the Null-Protein and Null-Compound models result in a Pearson correlation coefficient of ~ 0.6 in the PDBbind data set. The Null-Protein model and the Null-Compound model result in lower Pearson correlation coefficients in the Davis and Metz data sets than in the PDBbind data set, and this may be because Davis and Metz have higher completeness than PDBbind.

To evaluate the neural networks' predictive values on the interactions between unknown proteins and unknown compounds, we train the models on one data set and test them on another data set, for example, train on Metz/train and test on PDBbind/test. Compounds and proteins are dissimilar in the three data sets (Figure S3), so one data set can be used as an independent data set to test the predictive value of the models when training on another data set. When training on Davis/train and testing on Metz/test and PDBbind/test, the Pearson correlation coefficients of Yuel are 0.42 and 0.46, respectively, much higher than that of DeepConv-DTI (0.21, 0.10) and DeepDTA (0.21, 0.08); similarly, when training on Metz/train and testing on Davis/test and PDBbind/test, the Pearson correlation coefficients of Yuel are 0.33 and 0.40,

respectively, much higher than that of DeepConv-DTI (0.19 and 0.03) and DeepDTA (0.12 and 0.24). Thus, when training on Davis and Metz, Yuel has a higher generalizability than DeepConv-DTI and DeepDTA. When training on PDBbind, all three models result in low Pearson correlation coefficients, which may be caused by the low completeness of the PDBbind data set. The Null-Protein model and the Null-Compound model result in low Pearson correlation coefficients in all the independent tests. The testing results of all five models on all three data sets are displayed in Figures S4–S8.

Yuel Addresses the Overfitting Issue That Exists in Other Neural Network Models. To further evaluate the overfitting degree of the neural network models, we modify the test sets (Davis/test, Metz/test, PDBbind/test) by replacing each of the protein sequences with a single alanine, and we name the new test sets as Davis_A, Metz_A, and PDBbind_A. A single alanine is not supposed to have a strong binding with the compounds, and such test sets force the neural network models to predict the CPI by using only the compound information. However, we find that the performances of DeepConv-DTI and DeepDTA on Metz_A and PDBbind_A are close to that on Metz/test and PDBbind/test (Figure 3). For example, the Pearson correlation coefficients of DeepDTA on PDBbind/test and PDBbind_A are 0.77 (Figure 2) and 0.62 (Figure 3), respectively. On the contrary, Yuel's prediction performance falls drastically on Davis_A, Metz_A, and PDBbind_A (Figure 3), in line with expectations. An unbiased model should not yield a good correlation between the

predicted value and the experimental value if the prediction is only based on the compound. DeepConv-DTI and DeepDTA do yield high correlation values. Hence, these tools pose inherent biases due to overfitting. Of note, when training on the Davis data set, the overfitting issue in DeepConv-DTI and DeepDTA is alleviated (Figure 3), and this may be due to the high completeness of the Davis data set.

We speculate that the overfitting issue in DeepConv-DTI and DeepDTA is mainly caused by (1) the completeness of the training set and (2) the FC layers used in the encoding-FC paradigm. In these neural network models, the CPI is predicted by a function of the concatenated compound–protein feature matrix (eq 1)

$$P_{\text{aff}} = f(B) = f(WA) = f\left(\begin{bmatrix} W_{\text{CC}} & W_{\text{CP}} \\ W_{\text{PC}} & W_{\text{PP}} \end{bmatrix} \begin{bmatrix} A(C) \\ A(P) \end{bmatrix}\right)$$

$$= f\left(\begin{bmatrix} \omega_{11} & \cdots & \omega_{1k} & \omega_{1(k+1)} & \cdots & \omega_{1n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \omega_{k1} & \cdots & \omega_{kk} & \omega_{k(k+1)} & \cdots & \omega_{kn} \\ \omega_{(k+1)1} & \cdots & \omega_{(k+1)k} & \omega_{(k+1)(k+1)} & \cdots & \omega_{(k+1)n} \\ \cdots & \cdots & \cdots & \cdots & \cdots & \cdots \\ \omega_{n1} & \cdots & \omega_{nk} & \omega_{n(k+1)} & \cdots & \omega_{nn} \end{bmatrix} \begin{bmatrix} A_1 \\ \cdots \\ A_k \\ A_{k+1} \\ \cdots \\ A_n \end{bmatrix}\right) \quad (1)$$

where B is the concatenated compound–protein feature tensor with the shape (b, n, n, f) ; b is the batch size; k is the number of compound atoms; $n - k$ is the number of protein residues; f is the feature dimension; A is the concatenated compound–protein feature tensor before the FC layer with the shape (b, n, n, f) ; W is the weight matrix of the FC layer with the shape (n, n) ; $A(C)$ is the compound feature tensor before the FC layers with the shape (b, k, f) ; $A(P)$ is the protein feature matrix before the FC layer with the shape $(b, n - k, f)$; W_{CC} is the weight matrix for atom–atom interactions in the compound with the shape (k, k) ; W_{PP} is the weight matrix for residue–residue interactions in the protein with the shape $(n - k, n - k)$; W_{CP} and W_{PC} are the weight matrices for CPIs; ω_{ij} is the weight between A_i and A_j ; $A_1, A_2, \dots$, and A_k are the features of compound atoms before FC layers with the shape $(b, 1, f)$; and $A_{k+1}, A_{k+2}, \dots$, and A_n are the features of protein residues before FC layers with the shape $(b, 1, f)$. Importantly, the FC layer is applied directly to the concatenated compound–protein feature matrix, and the weight vector ω_{ij} of the FC layer is dependent on positions i and j of the compound atom ($i, j \leq k$) or protein residue ($i, j > k$). Thus, we refer to this kind of FC layer as a position-wise FC layer (Figure 4a).

In Yuel, the CPI is predicted by a function of the outer product of the compound feature tensor and the protein feature tensor (eq 2)

$$P_{\text{aff}} = f(B(C) \otimes B(P)) = f\left(\begin{bmatrix} B_1^T B_{k+1} & \cdots & B_1^T B_n \\ \cdots & \cdots & \cdots \\ B_k^T B_{k+1} & \cdots & B_k^T B_n \end{bmatrix}\right)$$

$$= f\left(\begin{bmatrix} (\omega(C)A_1)^T \omega(P)A_{k+1} & \cdots & (\omega(C)A_1)^T \omega(P)A_n \\ \cdots & \cdots & \cdots \\ (\omega(C)A_k)^T \omega(P)A_{k+1} & \cdots & (\omega(C)A_k)^T \omega(P)A_n \end{bmatrix}\right) \quad (2)$$

where P_{aff} is the predicted affinity; $B(C)$ is the final compound feature tensor with the shape (b, k, k, f) ; $B(P)$ is the final protein feature tensor with the shape $(b, n - k, n - k, f)$; $B_1, B_2,$

$\dots$, and B_k are the features of compound atoms with the shape $(b, 1, f)$; $B_{k+1}, B_{k+2}, \dots, B_n$ are the features of protein residues with the shape $(b, 1, f)$; $\omega(C)$ is the weight matrix for the compound with the shape (f, f) ; and $\omega(P)$ is the weight matrix for the protein with the shape (f, f) ; the outer product is calculated by treating the compound feature tensor and the protein feature tensor as vectors with each element as a vector of the features of a specific compound atom or protein residue. Crucially, the compound feature tensor ($B(C)$) and the protein feature tensor ($B(P)$) are processed by FC layers, that is, $\omega(C)$ and $\omega(P)$, separately. The weight vector $\omega(C)$ of the compound FC layers is not dependent on the position of compound atoms, and it is the same for all compound atoms; similarly, the weight vector $\omega(P)$ of the protein FC layers is the same for all protein residues. Thus, we refer to these FC layers as feature-wise FC layers (Figure 4b).

There are three advantages of feature-wise FC layers compared to position-wise FC layers. First, the weights in position-wise FC layers are dependent on the position of atoms and residues. However, when predicting CPI, the protein sequence may lack some residues, which changes the positions of residues, and the position of atoms in the compound may also be different if using a different encoding algorithm. Thus, the position-wise FC layers may result in different affinities for the same compound and the same protein. Second, most parameters of the weight matrix of the position-wise FC layers are focused on atom–atom interactions in the compound (W_{CC}) and residue–residue interactions in the protein (W_{PP}), while atom–residue interactions (W_{CP} and W_{PC}) are the critical ones for determining CPI; thus, the position-wise FC layers may make the network inclined to memorize the 2D structure of the small molecule and the sequence of the protein by optimizing W_{CC} and W_{PP} instead of W_{CP} and W_{PC} . Finally, since the weight matrix of the feature-wise FC layers is the same for all atoms or residues, it has far fewer parameters than the weight matrix of the position-wise FC layers, which makes the training of feature-wise FC layers much more efficient. In position-wise methods, the number of parameters in one FC layer is the size of the matrix $\begin{bmatrix} W_{\text{CC}} & W_{\text{CP}} \\ W_{\text{PC}} & W_{\text{PP}} \end{bmatrix}$, which is n^2 , where n

is the number of atoms in the compound plus the number of residues in the protein. In the feature-wise method, the number of parameters in one FC layer is the size of the matrix $\omega(C)$ plus the size of the matrix $\omega(P)$, which is $2f^2$, where f is the dimension of the feature. In Yuel, n is $(64 + 2048)$ and f is 16.

To test if there is a difference in performance between feature-wise versus position-wise FC layers, we create a new version of Yuel by using the position-wise FC layers to conform to the encoding-FC paradigm (Figure S9). We refer to this version of Yuel as Yuel-cc (concatenated version of Yuel). We train and test Yuel-cc on Davis and PDBbind data sets. We find that the Pearson correlation coefficients of Yuel-cc are 0.69 and 0.74 in the Davis and PDBbind data sets, respectively (Figure 4c). However, if we train Yuel-cc on Davis but test on PDBbind, the Pearson correlation coefficient is 0.10 (Figure 4c). Thus, by using the position-wise FC layers, Yuel-cc loses the generalizability to predict the interaction between unknown compounds and unknown proteins.

Identifying Hotspot Atoms and Hotspot Residues.

When encoding the protein feature matrix, each row corresponds to a residue, and each column corresponds to a specific feature. Similarly, each row of the compound feature

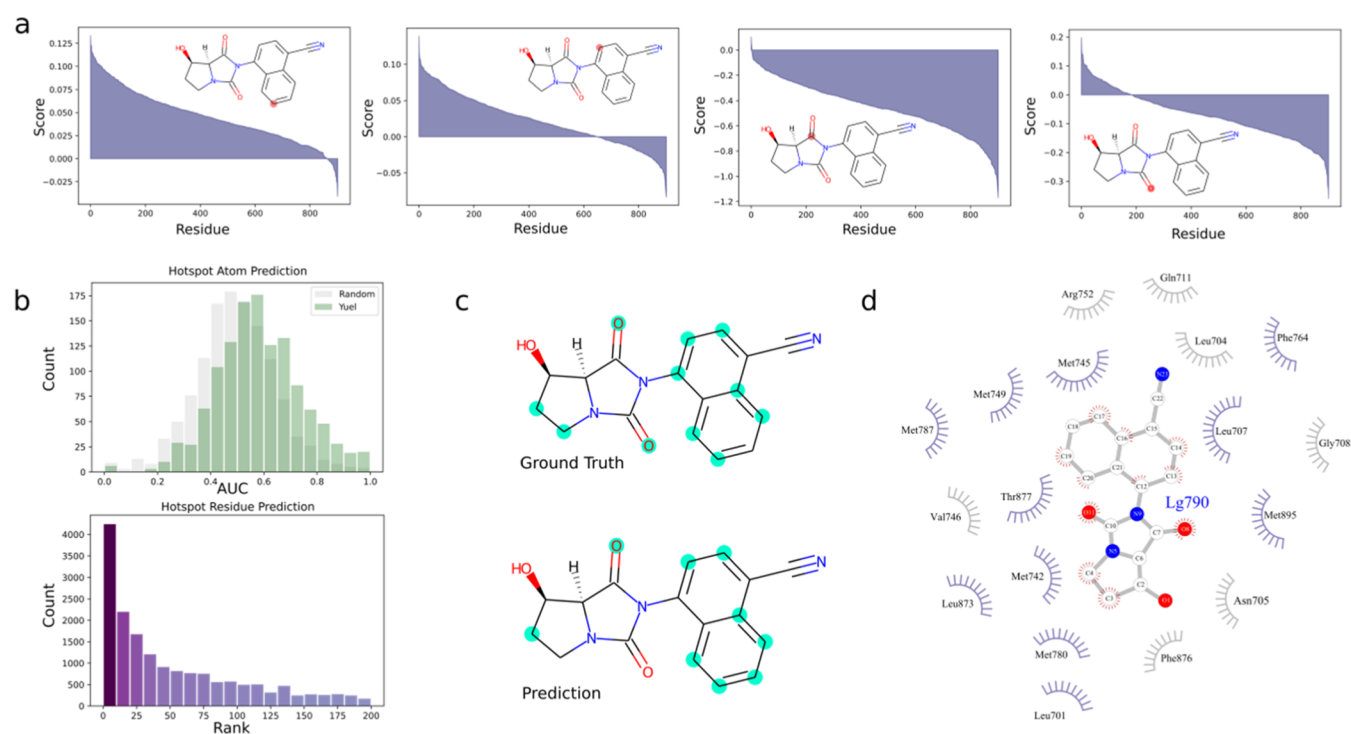


Figure 5. Testing the ability to identify hotspot atoms and residues. (a) Predicted protein-binding interactions of four atoms in *N*-aryl-hydroxybicyclohydantoin (LG790), the ligand of the rat androgen receptor (PDB ID: 2IHQ). The X-axis indicates the residues sorted by their scores from high to low. The Y-axis is the score of the compound atom corresponding to a protein residue. (b) Histogram of AUC of hotspot atom prediction in the PDBbind data set, and the histogram of the ranks of hotspot residues in the PDBbind data set. (c) Predicted and ground-truth hotspot atoms in LG790. (d) Predicted hotspot residues in the protein of 2IHQ.

matrix corresponds to an atom, and each column corresponds to a feature of the atom. After processing the two feature matrices by GNN, CNN, and feature-wise FC layers, Yuel calculates the outer product of the compound feature matrix by the protein feature matrix to obtain the attention matrix. In this attention matrix, each row corresponds to an atom in the compound, and each column corresponds to a residue in the protein. In this way, each element of the attention matrix corresponds to the interaction between an atom in the compound and a residue in the protein. After a sigmoid activation function, the resulting value can be interpreted as the probability that the corresponding atom and the corresponding residue can bind. Therefore, by comparing the values of different rows in the attention matrix, we can compare the probability of different compound atoms interacting with the protein so as to find the atoms most likely to bind the protein. We refer to atoms with particularly high interaction probabilities as hotspot atoms; similarly, we refer to residues that are most likely to interact with compounds as hotspot residues. Since different compound atoms have different predicted interactions with different residues (Figure 5a), we adopt a strategy of adding up the interactions of an atom to all residues as the possibility that this atom is a hotspot atom. This strategy results in an average area under the curve (AUC) (Methods) of 0.58 for PDBbind. The AUC distribution of all protein–ligand pairs (Figure 5b) shows that Yuel has more high AUC values and less low AUC values than a predictor that predicts the attention matrix randomly. Similarly, the predicted interactions of most hotspot residues are ranked within 30 (Figure 5b). Particularly, for 2IHQ, Yuel can predict 9 out of the 11 hotspot atoms (Figure 5c) and 11 out of the 18 hotspot residues (Figure 5d).

DISCUSSION

Many neural networks focus on the design of the encoder of the network to better extract protein and compound features. Here, we propose that the most important aspect in designing a CPI prediction network is learning the compound–protein binding patterns rather than learning the compounds and the proteins themselves to avoid overfitting to specific structures of the compounds alone or the proteins alone. We have demonstrated that the usage of position-wise FC layers is the culprit that leads to the low generalizability of neural network models, while the feature-wise FC layers in combination with the attention module significantly improve the generalizability of the network. In Yuel, after GNN layers, the feature of each compound atom contains the information of its neighboring atoms; similarly, after CNN layers, the feature of each residue contains information of its sequence-neighboring residues. In this manner, if a large value is obtained by multiplying the feature of a compound atom by the feature of a protein residue, then the group of atoms centered at this atom may interact with the fragment centered at this residue. The essence of Yuel is to predict the interaction between all possible combinations of atomic groups in the compound and amino acid fragments in the protein.

Neural networks can be used not only for virtual screening of drugs but also to identify protein targets, especially off-target proteins, but the premise is that the neural network must have high generalizability. For successful virtual screening, we must identify a small number of compounds which are most likely to bind the target protein out of an enormous library of compounds; similarly, for protein target identification, we must identify a single protein most likely to bind the specific compound from the proteome, which consists of millions of

protein species.^{32,33} For virtual screening, CPI prediction methods based on protein 3D structures are limited when there is no experimental structure of the protein; in this case, only the predicted protein structure can be used, which may compromise accuracy. In fact, the 3D structures of most proteins in the human genome have not been solved yet. Therefore, sequence-based CPI prediction methods have more extensive application capabilities, but because they are based on sequence rather than structure, these methods are easy to overfit, memorizing the input–output pair instead of trying to “understand” them and thus limiting the method’s power to generalize. Among sequence-based CPI prediction methods, Yuel has a higher generalizability than others by effectively addressing the overfitting problem.

In conclusion, we develop a new sequence-based neural network model, Yuel, to predict CPI with improved generalizability. In contrast to other neural network models, Yuel utilizes feature-wise FC layers and an attention module instead of subjecting the concatenated compound and protein feature matrices to position-wise FC layers. This new architecture endows Yuel with the ability to predict interactions between unknown compounds and unknown proteins.

METHODS

Architecture of Yuel. Yuel requires two inputs to estimate interactions between a specific compound and the protein target: (i) the compound’s 2D structure (SMILES²⁴ or INCHI³⁴ code) and (ii) the protein’s amino acid sequence. Given the protein sequence, Yuel first encodes it using the BLOSUM62 matrix³⁵ (<https://www.ncbi.nlm.nih.gov/Class/FieldGuide/BLOSUM62.txt>). Each residue corresponds to a column in the BLOSUM62 matrix, so each residue in a protein sequence is represented by a 20-element vector in Yuel. The features of non-standard amino acids are zero-initialized. Then, the protein features are updated through three one-dimensional (1-D) convolution layers. The sequence is zero-padded to a fixed length of 2048. Given the SMILES of the compound, Yuel first employs rdkit³⁶ to represent it by a graph (N, V, E), where N is the number of nodes, V is the feature vector of each atom, and E is the feature vector of each bond. The feature of each atom is the concatenation of the one-hot encoding of atom type, number of bonds, bond type, mass, and charge vectors. The feature of each bond is the bond order. The graph is then subject to two graph convolution layers to update the features. The protein features and the compound features are then subject to five feature-wise FC layers, separately. Following the FC layers, the protein feature matrix and the compound feature matrix are then multiplied to obtain an attention matrix. Each row of the attention matrix corresponds to a compound atom, and each column corresponds to a protein residue. The maximum value of each column is then selected and added up to get the final predicted CPI. We implement this network using TensorFlow 2.3.1³⁷ and Sonnet 2. The GNN is implemented by the *graph_nets*³⁸ library.

Hyper-Parameter Selection. The number of atom features is 82 by using a similar encoding method in Shuya Li’s work.²³ The first 63 atom features are the one-hot encoding of the atom element in the list of C, N, O, S, F, Si, P, Cl, Br, Mg, Na, Ca, Fe, As, Al, I, B, V, K, Tl, Yb, Sb, Sn, Ag, Pd, Co, Se, Ti, Zn, H, Li, Ge, Cu, Au, Ni, Cd, In, Mn, Zr, Cr, Pt, Hg, Pb, W, Ru, Nb, Re, Te, Rh, Tc, Ba, Bi, Hf, Mo, U, Sm, Os, Ir, Ce, Gd, Ga, Cs, and “unknown”; six features are the one-hot encoding of the degree of the atom; six features are the one-hot

encoding of the explicit valence of the atom; six features are the one-hot encoding of the implicit valence of the atom; and the last feature is whether the atom is aromatic. The number of bond features is 6, indicating whether the bond is single, double, triple, aromatic, or conjugated or in a ring. The number of residue features is 20, the number of amino acid types. The number of GNN layers is 2, and the number of CNN layers is 3. We choose 2 as the number of GNN layers because we consider each atom group as the center atom, the neighbor atoms of the center atom, and the neighbor atoms of the neighbor atoms. We choose 3 as the number of CNN layers such that each residue fragment contains seven residues. After the last CNN layer, we concatenate four protein feature matrices together: the protein feature matrix before the first CNN layer, the protein feature matrix after the first CNN layer, the protein feature matrix after the second CNN layer, and the protein feature matrix after the third CNN layer. Since our model only uses the heavy atoms in compounds, 64 as the number of atoms is enough to cover most compounds; similarly, 2048 as the number of residues is enough to cover most proteins. The size of the feature dimension is 16 for most layers in the model. Before multiplying the compound feature matrix and the protein feature matrix, we subject them to five FC layers, separately. The size of the feature dimension in the first four linear layers is 16×8 , and the size of the last linear layer is 16×50 .

Training. The objective is to minimize the mean squared error of the predicted binding affinity and the experimental binding affinity. During the training process, we use a mini-batch of 8 and the Adam stochastic optimizer³⁹ to optimize the model parameters. For each training batch, compounds with different numbers of atoms and proteins with different numbers of residues are zero-padded to obtain the same input feature lengths. During the training process, the padded regions of features are masked so that they do not contribute to the calculation of the losses and gradients. Yuel has about 2 million learnable parameters. A single Yuel model has 355 811 parameters if the “nfeat” variable is 16. We also train Yuel by setting the “nfeat” variable as 8, and the performance does not change significantly (Figure S10). Training Yuel by 200 epochs on a data set containing 440 000 CPIs needs 34 h on a Linux server with one NVIDIA Tesla T4 GPU.

Data Sets. Our benchmark data sets include the PDBbind data set,³¹ the Davis data set,²⁹ and the Metz data set.³⁰ The PDBbind data set is compiled from the PDBbind database (version 2018, the general set), which contains a high-quality set of protein–ligand complexes with available structural data and the corresponding binding affinities. Each complex was provided with an affinity value of a certain measurement type. The Davis data set is obtained from the Supporting Information of Davis et al. work.²⁹ The Metz data set is obtained from the Supporting Information of Metz et al. work.³⁰ The PDBbind data set contains 2509 small molecules and 10 251 proteins, with a total of 13 311 interactions. On average, each small molecule corresponds to only 5.3 interactions and each protein corresponds to only 1.3 interactions. The Davis data set contains 68 small molecules and 379 proteins, with a total of 9125 interactions. On average, each small molecule corresponds to 134.2 interactions, and each protein corresponds to 24.1 interactions. The Metz data set contains 1471 small molecules and 172 proteins, with a total of 35 307 interactions. On average, each small molecule corresponds to 24.0 interactions, and each protein corresponds

to 207.7 interactions. We divide each data set into a training set and a test set at a ratio of 8 to 2.

Comparison with Other Methods. The source codes of DeepConv-DTI are downloaded from <https://github.com/GIST-CSBL/DeepConv-DTI>. We implemented a new script to train and test with a custom data set. The source code of DeepDTA is downloaded from <https://github.com/hkmztrk/DeepDTA>. We have implemented a new script to train and test with a custom data set. We only compare Yuel to DeepConv-DTI and DeepDTA because MONN needs specific pair-wise interactions between atoms in compounds and residues in proteins for training, and DeepAffinity does not provide necessary utilities for training with custom data sets. All four models conform to the encoding-FC paradigm.

Evaluation of the Ability to Identify Hotspot Atoms and Residues. Since only the PDBbind data set provides the structural information, we evaluate the ability of Yuel to identify hotspot atoms and residues only in the PDBbind data set. For each of the compound–protein pair in the PDBbind test set, we employ Yuel to predict the binding affinity and extract the attention matrix. Each row of the attention matrix corresponds to an atom in the compound, and each column corresponds to a residue in the protein. Based on the attention matrix, we can rank the atoms in the compound by summing the columns for each row of the attention matrix, and we can rank the residues in the compound by summing the rows for each column of the attention matrix. Based on these predicted rankings and the ground truth, we can calculate true positives and false positives. Based on the true positives and false positives, we calculate the AUC of the receiver operating characteristic (ROC; also known as AUROC).

DATA AVAILABILITY

Source codes and test data are deposited in <https://bitbucket.org/dokhlab/yuel>. We provide a website to use Yuel: <https://yuel.dokhlab.org> or <https://dokhlab.med.psu.edu/cpi>.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c01531>.

Detailed testing results of Yuel and architectures of the Null-Protein, Null-Compound, and Yuel-cc models (PDF)

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Author Contributions

¹N.V.D. and J.W. designed and conceived the project. J.W. implemented the algorithm and performed the tests. J.W. and N.V.D. wrote the manuscript.

Notes

The authors declare no competing financial interest.

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